



Doping of carbon into boron nitride to get the increased adsorption ability for tetracycline from water by changing the pH of solution

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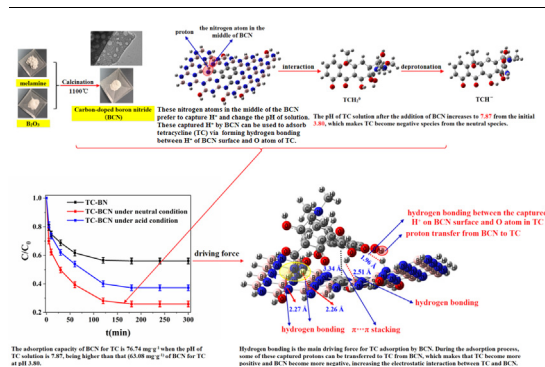
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HIGHLIGHTS

- Carbon-doped boron nitride (BCN) is synthesized via oxygen-limited method.
- BCN can capture protons and exist in protonated form in solution.
- BCN can increase the pH of tetracycline (TC) solution from 3.80 to 7.87.
- TC adsorption by BCN is 76.74 mg g^{-1} in pH 7.87, and 63.08 mg g^{-1} in pH 3.80.
- Some of the captured protons by BCN can transfer to TC during adsorption process.

GRAPHICAL ABSTRACT



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ABSTRACT

A series of carbon-doped boron nitride (BCN) are synthesized via oxygen-limited method, in which BCN-0.6 has the best adsorption quantity for tetracycline (TC). BCN-0.6 can capture protons and increase the pH of TC solution from 3.80 to 7.87, which makes that TC changes to negative state (pH 7.87) from neutral state (pH 3.80). BCN-0.6 exists in protonated form in solution. TC adsorption quantity by BCN-0.6 in pH 7.87 is higher than that in pH 3.8 (76.74 mg g^{-1} vs. 63.08 mg g^{-1}), being due to that these captured protons can also interact with these oxygen-containing groups in TC via hydrogen bonding. The hydroxyl numbers of BCN-0.6 have been determined via Boehm titration method for clarifying the contribution of these surface hydroxyl for TC adsorption. In addition, some of the captured protons by BCN-0.6 can transfer to TC during adsorption process, increasing the electrostatic interaction between BCN-0.6 and TC. Experimental and theoretical studies support that hydrogen bonding, electrostatic interaction and π - π stacking are main driving force for TC adsorption by BCN-0.6. Langmuir isotherm model and pseudo-second-order kinetic model are suitable to describe the adsorption of TC by BCN-0.6. And, this adsorption is an endothermic process. The adsorption performance of BCN-0.6 for TC is easily regenerated by washing with water. Furthermore, the per unit specific surface adsorption capacity (4.102 mg m^{-2}) of BCN-0.6 for TC is far higher than that of commercial activated carbon for TC (0.163 mg m^{-2}), suggesting that BCN-0.6 is a good adsorbent for water purification.

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1. Introduction

Tetracycline (TC), as a typical broad-spectrum antibiotic for treating and preventing infectious diseases of human and animals, has been widely used in human society, livestock husbandry and aquaculture [1–3]. However, residues of TC have been frequently detected in aquatic environment, including sewage, agricultural wastewater, surface water, ground water, and even drinking water, which has resulted in the threat to aquatic organisms and human [4–6]. Due to its antibiotic property and low biodegradability [2,7,8], it is not easy to remove TC from aqueous solution by biological method. So, how to high-efficiently remove TC from aquatic system has already become a major concern. Many methods have been adopted to remove TC from aqueous solution, such as advanced oxidation [9], electrochemistry [10], membrane filtration [11], and adsorption [2,3,12]. Among them, adsorption has been advised as a very competitive method due to the low cost, high efficiency and simple process [4,13]. A variety of different adsorbents including soils [14], clays [15], activated carbon [16] and porous synthetic resins [17] have been used to remove TC from aqueous solution. For instance, the adsorption capacity of magnetic hyper-crosslinked resins for TC is around $130 \text{ mg} \cdot \text{g}^{-1}$, and the specific surface area is $1322 \text{ m}^2 \cdot \text{g}^{-1}$ [9]. The adsorption quantity of activated carbon fiber heated by microwave for TC is around $300 \text{ mg} \cdot \text{g}^{-1}$, and the surface area is $1008 \text{ m}^2 \cdot \text{g}^{-1}$ [18].

During adsorption process, pH value of solution has a great influence on the adsorption performance of adsorbent for pollutants, being from that it can influence the surface potential of adsorbent and the existence form of the pollutants [19–21]. TC is an amphoteric molecule having multiple ionizable functional groups, including a carbonylamide group, a phenolic deketone group and a dimethyl amino group [22,23]. From Fig. 1, the three groups in TC can undergo protonation-deprotonation reactions and make TC exist as cation species TCH_3^+ ($\text{pH} < 3.3$),

zwitterions species TCH_2^0 ($3.3 < \text{pH} < 7.68$) or anion species TCH^- , TC^{2-} ($\text{pH} > 7.68$) in aqueous solutions [24]. So far, there are lots of reports about removing TC from water via adsorption method [9,14–18]. However, the researches about adsorbent-induced pH change of amphoteric pollutants solution (such as TC solution) and the resulted increase of adsorption quantity by adsorbent for the amphoteric pollutants are still not being found.

BCN is regarded as the material between carbon and boron nitride (BN) materials, and is a ternary hybrid material with different B, C, and N components [25–27]. Most of BCN materials are synthesized by using some organic compounds (such as glucose) as carbon sources [28,29]. And the researches by using melamine as both carbon source and nitrogen source are not too much. Except for the high chemical stability from BN, BCN also displays good adsorption ability, being due to that the conjugation between B (electron deficiency center) and N (rich electron center) is weakened by the doping of carbon into the BN skeleton [30–37]. It has been reported that the porous BCN had a higher

efficiency for removal of RhB than that of pure h-BN [34]. The synthesis of BCN mostly proceeds in an inert atmosphere [25–27], such as nitrogen gas. Analogous to BCN material, BN materials are also usually synthesized in inert atmosphere [29,31,32]. It has reported that BN materials calcinated in air atmosphere at 1000°C can be peeled into nanosheets with simultaneous hydroxylated functionalization. Our previous work shows that BN materials synthesized via one-step oxygen-limited method can directly produce hydroxyl groups on their surface [19,39]. Furthermore, BN and BN-based hybrid material can all change the pH of solution [19]. This encourages us to consider if BCN can be synthesized via oxygen-limited method and the synthesized BCN has interesting adsorption performance for amphoteric pollutants (such as TC). So far, BCN material has not been synthesized via oxygen-limited method, and its performance to change the pH of solution as well as its adsorption mechanism for amphoteric pollutants (such as TC) have not been reported as well.

Herein, a series of BCN materials have been first time successfully synthesized with diboron trioxide (boron source) and melamine (carbon source and nitride source) as precursors via oxygen-limited method and their adsorption performance for TC have been investigated as well. It is very interesting that BCN can increase the pH of TC solution from 7.87 to 3.80, which results in increased adsorption capacity of BCN for TC. The objectives of this work are included as the following: (1) synthesizing successfully BCN materials via oxygen-limited method; (2) clarifying the mechanism of BCN to change the pH of solution; (3) identifying the adsorption mechanisms of BCN material for TC.

2. Materials and methods

2.1. Materials and reagents

Diboron trioxide (B_2O_3 , analytical grade), melamine ($\text{C}_3\text{H}_6\text{N}_6$, analytical grade), hydrochloric acid (HCl, guaranteed reagent), sodium hydroxide (NaOH, analytical grade), sodium borohydride (NaBH_4 , analytical grade) and tetracycline (TC, analytical grade) were purchased from the Sinopharm Chemical Reagent limited corporation, P.R. China. All the reagents were used as received without further purification. The ultra-pure water was used throughout the experiment. The working solutions were prepared freshly.

2.2. Synthesis of BCN materials

6.962 g B_2O_3 and 6.306 g melamine with the molar ratio 1:0.5 were firstly dissolved in 100 mL of ultra-pure water. After that, the mixed solution was stirred and heated in a constant temperature water bath at 90°C for 8 h. Then, the mixture was dried in oven at 80°C for overnight to obtain the precursor. After grinded in mortar for half hour, the precursor was put into graphite crucible and covered with crucible lid. Finally, the crucible was put in the muffle furnace with the followed

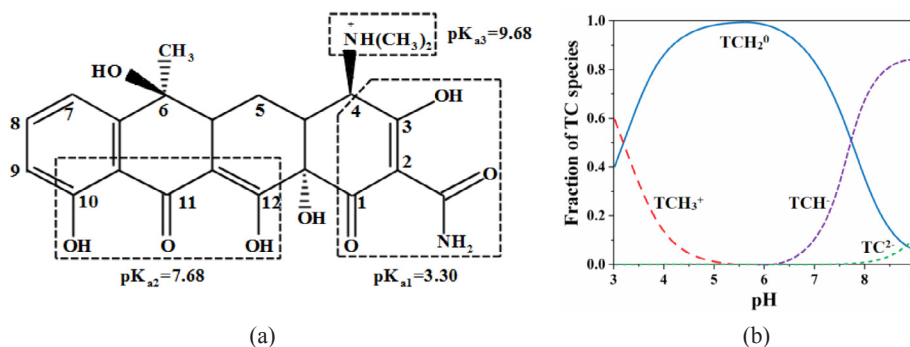


Fig. 1. (a) The structure of tetracycline (TC), the dotted areas represent the structural moieties associated with the three acidic dissociation constants (pK_a); (b) speciation distributions of TC in different pH values.

calcination at 1100 °C for 3 h. The heating rate is 10 °C/min. The obtained powder was washed with ultra pure water several times to remove the unreacted B₂O₃ and other impurities with the following drying in oven at 80 °C for overnight. This sample was defined as BCN-0.5 (Fig. S1a).

For considering the influence of melamine dosage on the adsorption performance of BCN composites, BCN-0.6, BCN-0.7, BCN-0.8 and BCN-0.9 composites were also produced in same procedure of synthesizing BCN-0.5 except for that the molar ratios between B₂O₃ and melamine were 1:0.6, 1:0.7, 1:0.8 and 1:0.9, respectively. The images of BCN-0.6, BCN-0.7, BCN-0.8 and BCN-0.9 samples were shown in Fig. S1b, Fig. S1c, Fig. S1d and Fig. S1e.

2.3. Characterization methods

The morphology of BCN-0.6 was investigated with Transmission electron microscope (TEM, JEM-200CX, Japan) at an acceleration voltage of 200 keV. The X-ray diffraction (XRD) patterns of BCN-0.6 was collected with the X-ray Diffractometer radiation (Ultima III Tokyo, Japan) using Cu K α ($\lambda = 1.540562 \text{ \AA}$) radiation source, in which the X-ray tube was operated at 40 kV and 40 mA. XPS characterizations of BCN-0.6 was performed with photoelectron spectroscopy (PHI 5000 VersaProbe, Japan). Specific surface area of BCN-0.6 was determined with surface analyzer (HD88, ASAP2020 micropore analyzer (USA)). The Fourier transform infrared spectrum (FTIR) of BCN-0.6 was acquired using infrared spectrometer (Nexus 870 FT-IR instrument, USA). Zeta potential of BCN-0.6 at different pH values was determined with potential analyzer (Zetasizer Nano ZS, UK). The pH values of the solutions were adjusted by adding 0.1 mmol and 0.01 mmol NaOH, or 0.1 mmol and 0.01 mmol HCl. The pH value of solution was measured by a pH meter (PHS-25, Shanghai yueping scientific instrument co., LTD, China).

The adsorption quantities of TC by BCN materials were determined via a UV-visible spectrometer (TU-1901, Beijing Purkinje General Instrument Co., Ltd, China) based on the change of their characteristic UV peaks before and after the adsorption experiments. According to Fig. S2, the characteristic UV peaks of TC solution at different PH values (7.87 vs. 3.80) have the similar height and are just little different in the adsorption wavelength: the characteristic UV peak of TC solution at pH 7.87 was 365 nm, while that of TC solution at pH 3.80 was 357 nm. In this work, the pH value of TC solution after addition of BCN-0.6 was changed to 7.87 from the initial 3.80. Thus, the standard curve of TC solution was drawn based on the characteristic UV peak of TC at 365 nm. From Fig. S3a, the UV peak (365 nm) of TC had good linear correlation with the concentration of TC since the correlation coefficient R^2 was 0.9997. From Fig. S3b, the UV peak (357 nm) of TC also had good linear correlation with the concentration of TC since the correlation coefficient R^2 was 0.9996, which was used to determine the adsorption capacity of BCN-0.6 for TC at pH 3.80.

2.4. The adsorption experiments of BCN materials for TC

2.4.1. The adsorption performance experiments

The adsorption performance of BCN-0.5, BCN-0.6, BCN-0.7, BCN-0.8 and BCN-0.9 were assessed by using them to adsorb TC. The detailed procedure was as following: 40 mg adsorbent was put into a 150 mL glass beaker with 100 mL TC solution (concentration: 40 mg·L⁻¹), followed by stirring at 200 rpm for 400 min at room temperature under dark condition. Then, this stirring was stopped for about ten minutes to make sure that the adsorbent was sinked to the bottom of the glass beaker. After that, 5 mL supernatant was taken from the beaker and passed through a 0.22 μ m cellulose membrane for the following characterization with UV-visible spectrometer to determine the adsorption quantities. From Fig. S4, the UV spectrum of TC solution did not change before and after filtration, which indicated that the cellulose membrane did not absorb TC. All of these adsorption

experiments were carried out three times to make sure that the obtained results were right.

2.4.2. The adsorption kinetic experiments

The adsorption kinetic of TC by BCN-0.6 was investigated in the same procedure mentioned above. For making sure that the obtained results were reasonable, eleven adsorption experiments of TC by BCN-0.6 had been done at the same time. Then, these experiments were stopped at 5 mins, 10 mins, 20 mins, 30 mins, 50 mins, 90 mins, 150 mins, 210 mins, 270 mins, 330 mins and 390 mins, respectively. 5 mL water samples were withdrawn and passed through a 0.22 μ m cellulose membrane for the following characterization with UV-visible spectrometer to determine the adsorption quantities of TC by BCN-0.6. These experiments had been performed three times to make sure that the obtained results were right.

In this study, the kinetics of the TC adsorption data were analyzed using the pseudo-first-order model, the pseudo-second-order model and the intra-particle diffusion model [40–42].

(1) The pseudo-first-order model

The pseudo-first-order equation was formulated as:

$$q_t = q_e(1 - e^{-k_1 t}) \quad (1)$$

where q_t (mg·g⁻¹) was the amount of TC adsorbed at time t (min), and q_e (mg·g⁻¹) represented the amount of TC adsorption at equilibrium, k_1 (min⁻¹) was the rate constant of pseudo-first-order adsorption.

(2) The pseudo-second-order model

The pseudo-second-order kinetic model was expressed by the following equation:

$$\frac{t}{q_t} = \frac{1}{k_2 q_e^2} + \frac{t}{q_e} \quad (2)$$

where k_2 (mg·g⁻¹·min⁻¹) was the adsorption rate constant. The parameters k_2 and q_e (mg·g⁻¹) were obtained from the intercept and slope of the plot of t/q_t versus t .

(3) The intra-particle diffusion model

The intra-particle diffusion model was explored by the following equation:

$$q_t = k_i t^{1/2} + C \quad (3)$$

where k_i (mg/g·min^{1/2}) was the intra-particle diffusion rate constant, and C (mg·g⁻¹) given an idea about the thickness of the boundary layer. The values of k_i and C were determined from the slope and intercept of q_t versus $t^{1/2}$ plotting.

2.4.3. The adsorption isotherm experiments

The adsorption isotherm of TC by BCN-0.6 was investigated in the same procedure mentioned above, except for that the concentration of TC was ranged from 30 mg L⁻¹, 40 mg L⁻¹, 50 mg L⁻¹, 60 mg L⁻¹, 70 mg L⁻¹, 80 mg L⁻¹, 90 mg L⁻¹ to 100 mg L⁻¹. All of these adsorption experiments were performed three times to make sure that the obtained results were right.

Three commonly used models, the Langmuir [41], Freundlich [43] and Temkin isotherm model [44], were applied to analyze the adsorption isotherm of TC by BCN-0.6.

(1) The Langmuir isotherm model

The Langmuir isotherm was based on the monolayer adsorption: the adsorbate distributed homogenously on the surface while no interaction existed between the adsorbate molecules [41]. The general Langmuir isotherm equation was described as follows:

$$\frac{C_e}{q_e} = \frac{1}{Q_m K_L} + \frac{1}{Q_m} C_e \quad (4)$$

where q_e (mg·g⁻¹) was the amount of adsorbate adsorbed under equilibrium; C_e (mg·L⁻¹) was the equilibrium concentration of TC; and Q_m (mg·g⁻¹) represented the theoretical maximum adsorption amount

adsorbed corresponding to the complete monolayer coverage; K_L ($\text{L}\cdot\text{mg}^{-1}$) was the Langmuir adsorption constant; Q_m and K_L were determined from the flowing linear form of the equation.

(2) The Freundlich isotherm model

The Freundlich isotherm model was employed to describe heterogeneous adsorbent surface and not restricted to monolayer formations, which was different from the Langmuir isotherm model [43]. The Freundlich isotherm equation was described by the following equation:

$$\ln q_e = \ln K_F + \frac{1}{n} \ln C_e \quad (5)$$

where q_e ($\text{mg}\cdot\text{g}^{-1}$) represented the equilibrium adsorption capacity; C_e ($\text{mg}\cdot\text{L}^{-1}$) was the equilibrium concentration of TC; K_F ($\text{mg}/\text{g}(\text{L}/\text{mg})^{1/n}$) was the empirical constant of adsorption capacity, and n represented the adsorption intensity, assessing the unfavorable adsorption ($n < 1$) or preferential adsorption ($n > 1$).

(3) The Temkin isotherm model

The Temkin isotherm model suggested that the heat of adsorption of all the molecules in the layer would decreased linearly with coverage owing to indirect adsorbent-adsorbate interactions and the binding energies distributed uniformly [44]. The Temkin isotherm in its linear form could be expressed:

$$q_e = B \ln K_T + B \ln C_e \quad (6)$$

where B was the Tempkin constant which related to heat of adsorption; K_T ($\text{L}\cdot\text{mg}^{-1}$) was the equilibrium binding constant; and B as well as K_T were determined from the slope and intercept of the plot.

2.4.4. The adsorption thermodynamic experiments

The adsorption thermodynamics of TC by BCN-0.6 was investigated in the same procedure mentioned above, except for that the reaction temperature was controlled as 25 °C, 35 °C, 45 °C and 55 °C. These experiments were performed three times to make sure that the obtained results were right.

The thermodynamic parameters, including the free energy changes (ΔG° ($\text{kJ}\cdot\text{mol}^{-1}$)), the standard enthalpy changes (ΔH° ($\text{kJ}\cdot\text{mol}^{-1}$)) and the entropy changes (ΔS° ($\text{kJ}\cdot\text{mol}^{-1}$)), were calculated to evaluate the exothermic nature of the adsorption process [45]. ΔG° was determined using the following equations:

$$\Delta G^\circ = -RT \ln(55.5 \times K_D) \quad (7)$$

where ΔG° ($\text{kJ}\cdot\text{mol}^{-1}$) represented Gibbs free energy; R ($\text{J}\cdot(\text{mol}\cdot\text{K})^{-1}$) represented the universal gas constant ($8.314 \text{ J}\cdot(\text{mol}\cdot\text{K})^{-1}$); T (K) represented absolute temperature; K_D represents equilibrium constant; The number 55.5 represented the amount of moles of H_2O per liter of solution. Equilibrium constant value of the adsorption was determined by the following equation:

$$K_D = \frac{q_e}{C_e} \quad (8)$$

The following equation was used to calculate ΔH° and ΔS° :

$$\ln(55.5 \times K_D) = \frac{-\Delta G^\circ}{RT} = \frac{-\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R} \quad (9)$$

2.5. The effect of the functional groups of BCN-0.6 on its adsorption for TC

300 mg BCN-0.6 was mixed with 300 mL deionized water and added into a 500 mL three-necked bottle. 3 g NaBH_4 was added into the bottle and reacted under stirring at 98 °C for 8 h in a water bath. Then, the precipitation was centrifuged to retain and washed with deionized water for several times with following drying in oven at 80 °C for overnight. The sample was marked as RBCN-0.6.

2.6. Recycle experiments of BCN-0.6 for TC

The adsorption experiments were carried out in the same procedure mentioned above. After that, the sample collected by the centrifuge was washed several times with ultra-pure water and dried at 80 °C for overnight. Then, the treated BCN-0.6 material was reused to adsorb TC solution (concentration: 40 mg L^{-1}) in the same procedure. For ensuring the accuracy of the experimental results, each experiment was repeated three times with three parallel samples each time.

2.7. The adsorption experiments of BN, biochar, activated carbon and LBCN-0.6

In order to evaluate the adsorption performance of BCN-0.6, the adsorption of TC by BN, biochar activated carbon and LBCN-0.6 were done as well. BN material was synthesized by calcinating B_2O_3 and urea under 1100 °C for 4 h with oxygen limited method in our previous work [46]. The heating rate was $10 \text{ }^\circ\text{C}\cdot\text{min}^{-1}$. Biochar was synthesized by heating the wheat straw powder in a furnace at 450 °C via oxygen limited method in our previous work [21]. LBCN-0.6 sample with large surface area was synthesized with the same procedure to synthesize BCN-0.6 except for adding NH_4CO_3 as pore-forming agent. Activated carbon was purchased from Hengsheng environmental protection material co., LTD.

A dose of adsorbents (40 mg), including BN, biochar, activated carbon and LBCN-0.6, were put into three glass beaker respectively, which contained 100 mL TC solution with the concentration ($40 \text{ mg}\cdot\text{L}^{-1}$). Then the solution was stirring at 200 rpm at room temperature under dark condition until the adsorption equilibrium was reached. After that, 5 mL supernatant was collected and passed through a $0.22 \mu\text{m}$ cellulose membrane for the following measurement of the adsorption quantity of adsorbents for TC by a UV-visible spectrophotometer. All the experiments was performed three times to make sure that the obtained results were right.

2.8. Computational section

The molecular structures of BN and BCN-0.6 as well as the adsorption performance of BCN-0.6 had been investigated with m062x/6–31 g(d, p) method 36 [47] in Gaussian 09 program [48]. The BN model was built with 36 boron atoms, 36 nitrogen atoms and 5 oxygen atoms, and the edges of BN model were saturated with 24 hydrogen atoms (Fig. S5(a)). The BCN model was constructed with 31 boron atoms, 8 carbon atoms, 33 nitrogen atoms and 5 oxygen atoms and the edges of BCN-0.6 were saturated with 24 hydrogen atoms (Fig. S5(b)). Furthermore, the interaction energy between BN, BCN-0.6 and protons as well as the adsorption energy of BCN-0.6 for TC were also investigated. The molecular structure of the negative TC at pH 7.87 was built, in which a proton is lost from the O atom and another proton was lost from the N atom (please see the two green circles in Fig. S5(c)). All molecular structures were constructed with Gview 5.0 program based on these optimized results.

3. Results and discussion

3.1. Determining the optimum molar ratio between diboron trioxide and melamine in BCN samples

The adsorption performance of BCN samples are first evaluated in this work to determine the optimum molar ratio between diboron trioxide and melamine. TC is a typical antibiotic, which has been widely used to evaluate the adsorptive performance of the synthesized materials [19,36,37,49–51]. Herein, TC is adopted to evaluate the adsorption performance of BCN samples.

From Fig. 2b, BCN-0.6 has the largest adsorption quantity (76.74 mg g^{-1}) for TC, followed by BCN-0.5 (68.48 mg g^{-1}), BCN-0.7

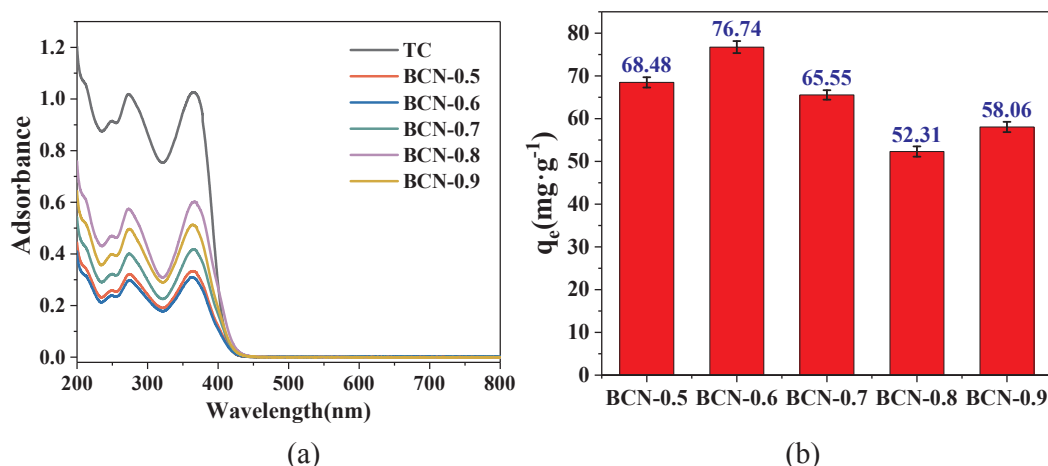


Fig. 2. (a) the UV-visible spectra of TC adsorption by BCN samples for 400 min (dosage: 40 mg, TC concentration: 40 mg L⁻¹, volume: 100 mL); (b) the adsorption quantities of TC by BCN samples.

(65.55 mg g⁻¹), BCN-0.9 (58.06 mg g⁻¹) and BCN-0.8 (52.31 mg g⁻¹). For understanding the adsorption performance difference of BCN-0.5, BCN-0.6, BCN-0.7, BCN-0.8 and BCN-0.9, the BET surface areas, zeta potential and surface hydroxyl groups of these BCN samples are investigated. According to the Table S1, BCN-0.6 has the largest surface area (18.708 m²·g⁻¹). This can explain well why BCN-0.6 has the largest adsorption quantity for TC (76.74 mg g⁻¹). But, BCN-0.8 has the smallest adsorption quantity (52.31 mg g⁻¹) for TC although the surface area (10.832 m²·g⁻¹) of BCN-0.8 is larger than that of BCN-0.9 (10.124 m²·g⁻¹), BCN-0.7 (8.855 m²·g⁻¹) and BCN-0.5 (7.501 m²·g⁻¹). Thus, surface area is not the main driving force for TC adsorption by BCN samples. It is true for the zeta potential since the order of TC adsorption quantity by BCN samples is not same as that of zeta potential on these BCN samples surface. So, zeta potential is not the main driving force for TC adsorption by BCN samples as well.

From Fig. 1a, TC has many functional groups, such as OH groups, NH₂ groups etc. Based on the FTIR spectra of BCN-0.6 in Fig. 3a, a strong O-H stretching vibration peak (3400 cm⁻¹) is found, indicating that there are OH groups on the surface of BCN-0.6. So, the interaction between the OH groups of BCN-0.6 and these functional groups (OH groups, NH₂ groups, etc) of TC may make contribution for TC adsorption by BCN-0.6. For confirming this point, the quantities of OH groups on BCN samples surface are determined via Boehm titration method. From Table S1, BCN-0.6 has the most OH groups (0.208 mmol·g⁻¹), followed by BCN-0.5 (0.198 mmol·g⁻¹), BCN-0.7 (0.184 mmol·g⁻¹), BCN-0.9 (0.143 mmol·g⁻¹) and BCN-0.8 (0.126 mmol·g⁻¹). This is consistent with the adsorption quantity order of TC by BCN samples:

BCN-0.6 has the largest adsorption quantity (76.74 mg·g⁻¹) for TC, followed by BCN-0.5 (68.48 mg·g⁻¹), BCN-0.7 (65.55 mg·g⁻¹), BCN-0.9 (58.06 mg·g⁻¹) and BCN-0.8 (52.31 mg·g⁻¹). Thus, the interaction between the OH groups of BCN-0.6 and these functional groups (OH groups, NH₂ groups, etc) of TC is the main driving force for the adsorption of TC by BCN samples, which make the adsorption difference of BCN-0.5, BCN-0.6, BCN-0.7, BCN-0.8 and BCN-0.9 for TC. Thus, BCN-0.6 is chosen for further investigating its structure and adsorption mechanisms for TC since it has the best adsorption performance for TC.

3.2. Characterizations of BCN-0.6

The FTIR spectrum of BCN-0.6 is given in Fig. 3a. The FTIR peak at 786 cm⁻¹ and the broad peak around 1466 cm⁻¹ are attributed to the bending vibration of the B-N-B bond and the stretching vibration of the B-N bond [38,39], respectively. The peak at 921 cm⁻¹ is assigned to N-B-O bond [52-55]. In addition, the peak at 2369 cm⁻¹ is ascribed to the C-N bending vibration, which supports that carbon from melamine has been successfully doped into the BN skeleton [52,56]. The broad peak at 3400 cm⁻¹ is assigned to the stretching vibration from the surface O-H bond and N-H bond [52,57]. Thus, there are OH groups and NH groups on the surface of BCN-0.6 when it is synthesized via oxygen-limited method.

The XRD spectrum of BCN-0.6 is shown in Fig. 3b, which have two main characteristic diffraction peaks at 26.7° and 42.5°, revealing the (0 0 2) and (1 0 0) crystal reflections of BCN material [58]. BN has two typical XRD peaks at 26.7° and 41.6° [59]. The observed shifts might be

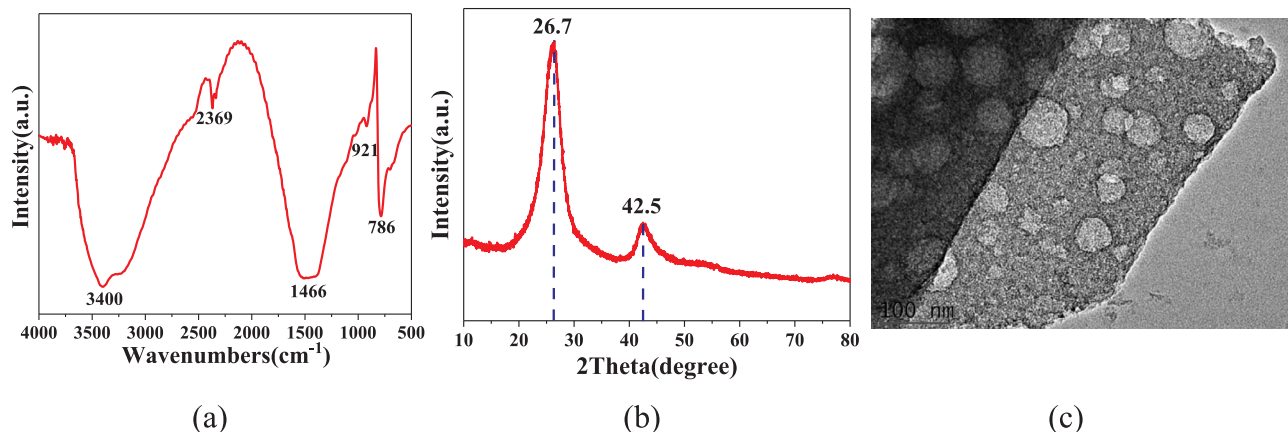


Fig. 3. (a) FTIR spectra of BCN-0.6; (b) XRD spectra of BCN-0.6; (c) TEM image of BCN-0.6, and the scale is 100 nm.

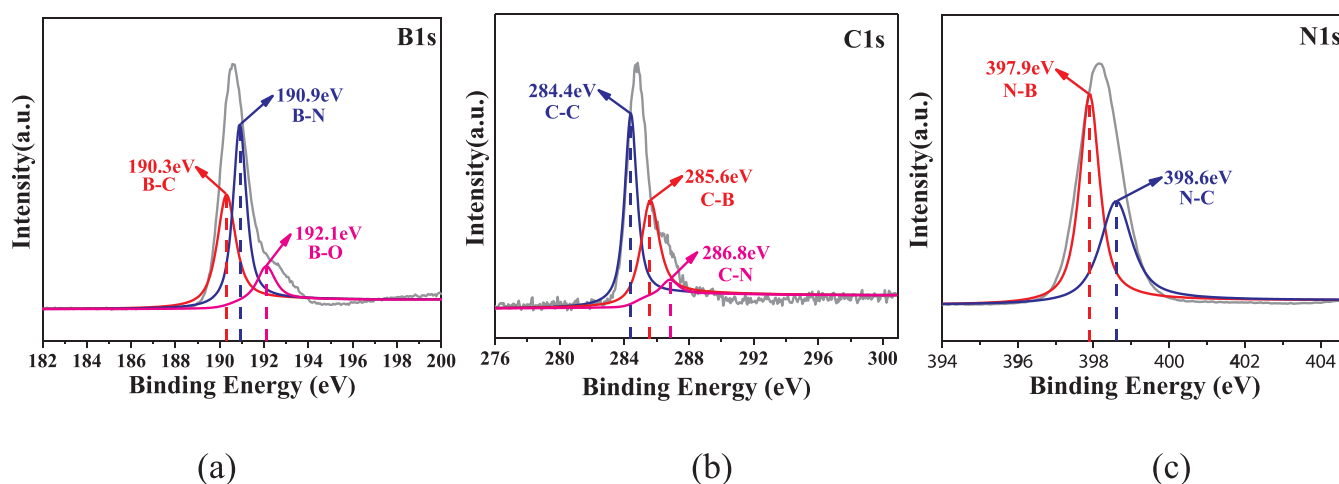


Fig. 4. (a) B1s XPS spectra of BCN-0.6 (b) C1s XPS spectra of BCN-0.6; (c) N1s XPS spectra of BCN-0.6.

related to the incorporation of carbon into the BN skeleton [31,32,35,52,58–60]. From Fig. 3c, BCN-0.6 is of a plate structure with circle hole on its surface [61,62]. This circle hole is most likely from the combustion of some of melamine since melamine contains C and N elements.

XPS is an effective way to characterize the electronic structure of element, which can be used to investigate the interactions between elements. The B1s, C1s, and N1s XPS spectra of BCN-0.6 are shown in Fig. 4. From Fig. 4a, The B 1s peak can be fitted into three sub-peaks at 190.3 eV, 190.9 eV and 192.1 eV corresponding to B-C, B-N and B-O bonds [26,59]. The C 1s signal (Fig. 4b) can be fitted into sub-peaks centered at 284.4 eV, 285.6 eV, and 286.8 eV, which are due to sp^2 carbons of C-C, C-B, and C-N bonds, respectively [28,29,60]. The N1s signal (Fig. 4c) at 397.9 eV and 398.6 eV is due to N-B and N-C bonds, respectively [26,52,63]. Thus, the XPS results support that carbon element has been successfully doped into the BN skeleton. From Fig. S1, the colors of BCN-0.5, BCN-0.6, BCN-0.7, BCN-0.8 and BCN-0.9 change from white to black, which is also a good evidence to support the successful doping of carbon into the BN skeleton and melamine can work as both N source and C source in this experiment.

3.3. Adsorption kinetic and adsorption isotherm

The adsorption mechanism of BCN-0.6 for TC is further investigated. From Fig. 5a, the adsorption quantity of BCN-0.6 for TC is up to $72.11 \text{ mg}\cdot\text{g}^{-1}$ in 150 min, then followed by the arrival of

adsorption equilibrium with the adsorption quantity ($74.12 \text{ mg}\cdot\text{g}^{-1}$) when the adsorption time is $> 210 \text{ min}$. According to Fig. S6, Table S2 and Table S3, the pseudo-second-order model has highest correlation coefficients ($R^2 = 0.9998$), followed by the intra-particle model (0.9527) and pseudo-first-order model (0.9257). The experimental and calculated values of q_e are very close, indicating that pseudo-second-order kinetic model is more suitable for describing the adsorption kinetic of TC by BCN-0.6 than other two models [45].

According to Fig. 5b, the adsorption quantity of TC by BCN-0.6 increases to $111.52 \text{ mg}\cdot\text{g}^{-1}$ from $65.47 \text{ mg}\cdot\text{g}^{-1}$ when the concentration of TC ranges from $30 \text{ mg}\cdot\text{L}^{-1}$ to $100 \text{ mg}\cdot\text{L}^{-1}$. From Fig. S7 and Table S4, the R^2 from Langmuir, Freundlich and Temkin isotherm models are 0.9992, 0.9609 and 0.9759, respectively. Thus, Langmuir isotherm model is more suitable for describing the adsorption isotherm process than the other two models [22,42].

3.4. Adsorption thermodynamics

The relationship between adsorption quantity and temperature have also been investigated as well. From Fig. 5c, it is obviously to see that the adsorption quantity increased to $85.62 \text{ mg}\cdot\text{g}^{-1}$ from $76.74 \text{ mg}\cdot\text{g}^{-1}$ when the temperature increases to 55°C from 25°C . This suggests that the adsorption of TC by BCN-0.6 is an endothermic process. Furthermore, the thermodynamics parameters for TC adsorption by BCN-0.6 are also calculated. According to Fig. S8 and Table S5, the calculated ΔG° at different temperatures are all negative, indicating that the TC

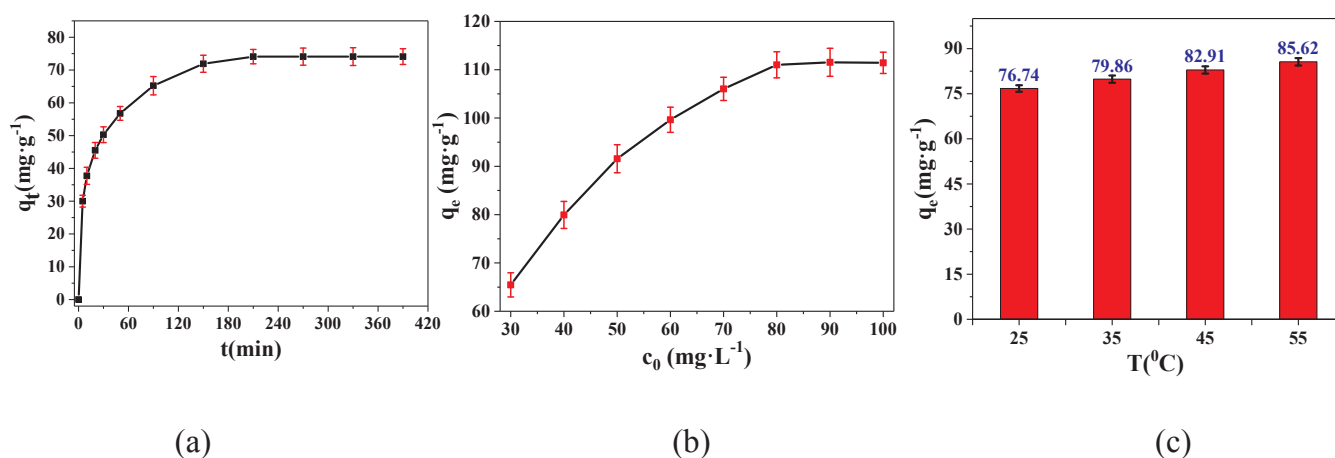


Fig. 5. (a) Kinetic adsorption of TC by BCN-0.6 (dosage: 40 mg, TC concentration: $40 \text{ mg}\cdot\text{L}^{-1}$, volume: 100 mL); (b) the isotherm adsorption curve of TC by BCN-0.6 (dosage: 40 mg, TC concentrations ranging from $30 \text{ mg}\cdot\text{L}^{-1}$ to $100 \text{ mg}\cdot\text{L}^{-1}$, volume: 100 mL); (c) The adsorption quantity of TC by BCN-0.6 at different temperatures.

adsorption by BCN-0.6 material was spontaneous [6]. With the increasing of temperature, the absolute values of ΔG° increase, which implies that higher temperature is favorable for the TC adsorption by BCN-0.6 [64]. The value of ΔH° ($16.03 \text{ kJ}\cdot\text{mol}^{-1}$) is positive for TC, suggesting that the reaction is endothermic. The value of ΔS° ($0.11 \text{ kJ}\cdot\text{mol}^{-1}$) calculated from the equation is positive, suggesting that the adsorption is spontaneous and favorable [20].

3.5. Study on the adsorption mechanism of TC by BCN-0.6

In order to confirm whether surface area is the adsorption driving force of BCN-0.6 for TC, the relationship between the adsorption quantity and BET surface of all BCN samples are investigated. From Fig. 2, the adsorption quantity of BCN-0.6 for TC is 76.74 mg g^{-1} , followed by BCN-0.5 (68.48 mg g^{-1}), BCN-0.7 (65.55 mg g^{-1}), BCN-0.9 (58.06 mg g^{-1}) and BCN-0.8 (52.31 mg g^{-1}). From Table S1 and Table S6, the BET surface area of BCN-0.6 is $18.708 \text{ m}^2\cdot\text{g}^{-1}$, followed by BCN-0.8 ($10.832 \text{ m}^2\cdot\text{g}^{-1}$), BCN-0.9 ($10.124 \text{ m}^2\cdot\text{g}^{-1}$), BCN-0.7 ($8.855 \text{ m}^2\cdot\text{g}^{-1}$) and BCN-0.5 ($7.501 \text{ m}^2\cdot\text{g}^{-1}$). Clearly, the adsorption quantity has not direct correlation with the BET surface area. Thus, BET surface is not the main driving force for the TC adsorption by BCN-0.6.

The pH value of TC solution is 3.80 at the beginning time of the experiment, TC exists in both form of neutral ions (TCH_2^0), and cations (TCH_3^+) (please see Fig. 1b). The pH of TC solution changes to 7.87 in 2 min after the addition of BCN-0.6 into TC solution, which leads to that TC loses the proton and becomes the negative species (TCH^-) (please see Fig. 1b). The surface electrostatic potentials of BCN-0.6 are -22 mV when the pH of solution is 3.8, while it is -28 mV when the pH of solution is 7.8. If the adsorption of TC by BCN-0.6 is driven by electrostatic interaction. The adsorption capacity of BCN-0.6 for TC in acidic condition (pH 3.80) should be larger than that in neutral condition (pH 7.87) since BCN-0.6 has a negative surface and TC exists in both neutral and positive forms when the pH of solution is 3.8. However, the adsorption capacity of BCN-0.6 for TC is 76.74 mg g^{-1} when the pH of the solution is 7.8, while the adsorption capacity of BCN-0.6 for TC is 63.08 mg g^{-1} when the pH of solution is 3.8. It seems that the electrostatic interaction is also not the main driving force for the TC adsorption by BCN-0.6. This can be further confirmed by the TC adsorption by BN. The surface electrical property of BN is -27 eV , which is similar to that of BCN-0.6. And its adsorption capacity for TC is 43.96 mg g^{-1} , which is much lower than that of BCN-0.6 (76.74 mg g^{-1}), supporting that electrostatic interaction is not the main driving force for the adsorption of TC by BCN-0.6.

For completely understanding the above adsorption results in different pH values and why BCN-0.6 can increase the pH of the TC solution, detailed theoretical studies have been done to investigate the interaction of BCN-0.6 with proton and TC. The molecular structures of BN model and BCN model are shown in Fig. S5. Protons may have three way to interact with BCN model (please see Fig. 6): one is that the two protons combine with the two middle N atoms (Fig. 6a), the calculated interaction energy is $331.5 \text{ kcal}\cdot\text{mol}^{-1}$; the second one is that the two protons combine with the two edge N atoms (Fig. 6b), the interaction

energy is $328.8 \text{ kcal}\cdot\text{mol}^{-1}$; the third one is that one proton combines with an edge N atom and another proton combines with an edge O atom (Fig. 6c), the interaction energy is $301.6 \text{ kcal}\cdot\text{mol}^{-1}$. Thus, protons prefer to combine with the middle N atoms in the BCN model and the high interaction energy of BCN model for proton ($331.5 \text{ kcal}\cdot\text{mol}^{-1}$) also explain well why BCN-0.6 can increase the pH of TC solution from 3.80 to 7.87. This is from that the protons in TC solution are captured by BNC-0.6. So, BCN-0.6 exists in the protonated form in the TC solution. As we know, this is first time to report that BCN exists in the protonated way during adsorption process. This finding can explain well why the surface electrostatic potentials of BCN-0.6 are -22 mV when the pH of solution is 3.8, while it becomes -28 mV when the pH of solution is 7.8. This is due to that some of the captured protons by BCN-0.6 are released when the pH of solution changes to 7.87 from 3.80.

For deep understanding the effect of doping carbon into BN skeleton, the interaction energies between BN model and protons are also calculated. According Fig. S9, the calculated interaction energy is $277.8 \text{ kcal}\cdot\text{mol}^{-1}$ when the two protons combine with the two middle N atoms (Fig. S9a); the calculated interaction energy is $318.6 \text{ kcal}\cdot\text{mol}^{-1}$ when the two protons combine with the two edge N atoms (Fig. S9b); the interaction energy is $304.1 \text{ kcal}\cdot\text{mol}^{-1}$ when one proton combines with an edge N atom and another proton combines with an edge O atom (Fig. S9c). Thus, BN can also change the pH of solution by capturing proton and it prefers to capture protons by using its edge N atoms. This is different from that of BCN-0.6, being from the following reason: doping of carbon into BN skeleton weakens the conjugation between B atom and N atom, and makes that the middle N atoms are more easily to share its electron with proton than that in BN. The HOMO and LUMO orbitals of BN and BCN models are given in Fig. S10 and Fig. S11. In BN model, it is clearly to see that the orbitals of B atom and N atom are conjugated together in both HOMO and LUMO orbitals (Fig. S10). And for BCN model, the doping carbon atoms participate the conjugation in both HOMO and LUMO orbitals (Fig. S11). This kinds of participation may make that N element in BCN is more easily to share its lone electron pair with proton than that of BN. Furthermore, this calculated interaction energy between BN and proton ($318.6 \text{ kcal}\cdot\text{mol}^{-1}$) is smaller than that between BCN and proton ($331.5 \text{ kcal}\cdot\text{mol}^{-1}$). This explains well why the ability of BCN to increase the pH of TC solution (from 3.80 to 7.87) is stronger than that of BN (from 3.80 to 4.12). Similar to BCN, BN also exists in the protonated form in solution during adsorption process.

From FTIR results (Fig. 3a), there are functional groups (OH and NH groups) on the surface of BCN-0.6 since it is synthesized via oxygen-limited method. TC also has functional groups (OH and NH_2 groups) (Fig. 1a). Thus, the interaction between the functional groups (OH and NH groups) in BCN-0.6 and the oxygen-containing groups (OH and NH_2 groups) in TC maybe make the main contribution for TC adsorption by BCN-0.6. Theoretical studies have been adopted to investigate the interaction between the functional groups (OH and NH groups) in BCN-0.6 and the oxygen-containing groups (OH and NH_2 groups) in TC. Herein, BCN model takes the protonated form since the above

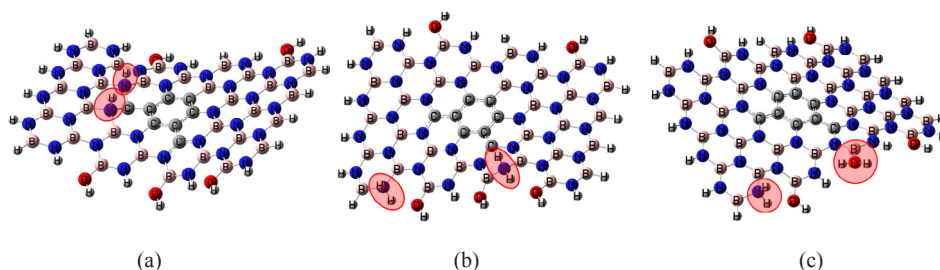


Fig. 6. (a) Two protons combine with the two middle nitrogen atoms in BCN model; (b) two protons are combined with the two edge nitrogen atoms in BCN model; (c) one proton is combined with the edge nitrogen atom in BCN model and another proton is combined with the edge oxygen atom in BCN model.

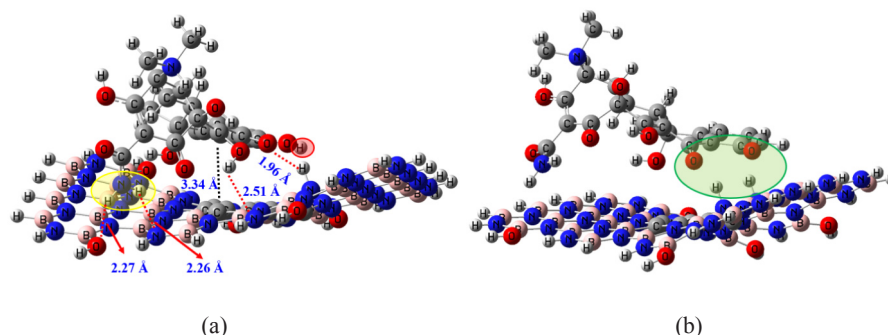


Fig. 7. (a) The structure of TC-BCN model after optimization; (b) the structure of TC-BCN before optimization.

experiment and theoretical studies all support that BCN-0.6 exists in the protonated form in solution (please see Fig. 6a). For TC, a negative form (TC^-) is taken since the pH of TC solution is 7.87 after the addition of BCN-0.6. The optimized structure of TC-BCN model is shown in Fig. 7, and the adsorption energy between BCN model and TC is calculated as $66.02 \text{ kcal}\cdot\text{mol}^{-1}$. From Fig. 7a, the distance between the C-bound O atom in TC and the N-bound proton in BCN is $1.96 \text{ \AA}$, the distance between the O-bound H atom in TC and the N atom in BCN-0.6 is $2.51 \text{ \AA}$, the distance between the N-bound H atoms in TC and the O atom, N atom in BCN-0.6 are $2.27 \text{ \AA}$ and $2.26 \text{ \AA}$, respectively. All of these suggest that the hydrogen bonding interaction between the functional groups (OH and NH groups) in BCN-0.6 and the oxygen-containing groups (OH and NH_2 groups) in TC make significant contribution for the TC adsorption by BCN-0.6. Another interesting insight from the theoretical results is that the proton on the surface of BCN model can transfer to the O atom in the negative TC that lost the proton (please see two circles in Fig. 7a and b and Fig. S5c). The kind transfer will make TC become positive and BCN model become negative, which will increase the electrostatic interaction between BCN-0.6 and TC, and enhance the TC adsorption by BCN-0.6. In addition, the distance between the aromatic ring in TC and the surface BCN model is $3.34 \text{ \AA}$, suggesting that $\pi\cdots\pi$ stacking also exists between TC and BCN model. Thus, hydrogen bonding between functional groups of BCN-0.6 and TC, the electrostatic interaction from the proton transfer from BCN-0.6 to TC, as well as $\pi\cdots\pi$ stacking between BCN-0.6 and TC plane all make the contribution for TC adsorption by BCN-0.6. In summary, doping of carbon element into BN skeleton makes that BCN-0.6 can capture more H^+ , has stronger ability to change the pH of TC solution than BN and change the existing form of TC. These captured H^+ by BCN-0.6 can further work as adsorption sites for TC. All of this makes that the adsorption quantity ($76.74 \text{ mg}\cdot\text{g}^{-1}$) of TC by BCN-0.6 in pH 7.87 is higher than that ($63.08 \text{ mg}\cdot\text{g}^{-1}$) of BCN-0.6 for TC in pH 3.80 and that ($43.96 \text{ mg}\cdot\text{g}^{-1}$) of BN for TC.

For confirming that the interaction between functional groups of BCN-0.6 and TC makes main contribution for the TC adsorption by BCN-0.6, NaBH_4 has been adopted to treat BCN-0.6 for introducing more OH functional groups on the surface of BCN-0.6 [65,66], and this treated sample is named as RBCN-0.6. From the FTIR spectra of BCN-0.6 and RBCN-0.6 (Fig. 8), the O–H vibration absorption peak intensity of RBCN-0.6 at the wavelength of 3400 cm^{-1} is significantly enhanced. The hydroxyl number of BCN-0.6 and RBCN-0.6 are determined via Boehm titration method [67,68], which are $0.208 \text{ mmol}\cdot\text{g}^{-1}$ and $0.324 \text{ mmol}\cdot\text{g}^{-1}$, respectively. Therefore, the O–H functional groups are introduced on the BCN-0.6 surface by NaBH_4 reduction [65,66]. The adsorption capacity of RBCN-0.6 for TC was $88.21 \text{ mg}\cdot\text{g}^{-1}$, which is increased by $11.47 \text{ mg}\cdot\text{g}^{-1}$ compared with BCN-0.6 ($76.74 \text{ mg}\cdot\text{g}^{-1}$). The zeta potential of RBCN-0.6 was still negative (-26 mV) and its BET surface area did not change ($18.7 \text{ m}^2\cdot\text{g}^{-1}$). All of these support that the interaction between the functional groups (OH and NH groups) in BCN-0.6 and the oxygen-containing groups (OH and NH_2 groups) in TC is the

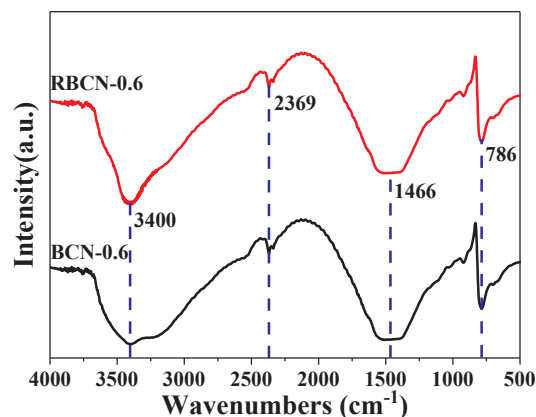


Fig. 8. The FTIR spectra of BCN-0.6 and RBCN-0.6.

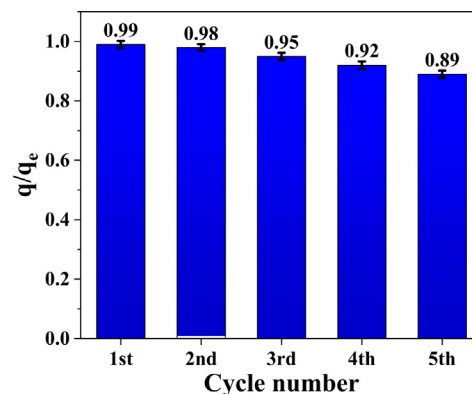


Fig. 9. The recycle experiment of TC removal by BCN-0.6.

main driving force for the TC adsorption by BCN-0.6.

3.6. BCN-0.6 recycle performance

The recycle performance of BCN-0.6 for TC adsorption is also investigated. From Fig. 9, the values of the ratio are 99% in the first time, 98% in the second time, 95% in the third time, 92% in the fourth time and 89% in the fifth time, respectively. A slight downward trend in the regeneration performance in the cycles. This may be attributed to that some TC have been adsorbed too tightly and are not easily washed off, which result in that the adsorption sites on the surface of BCN-0.6 decrease. Thus, the adsorption performance of BCN-0.6 for TC adsorption can be effectively regenerated by washing with water.

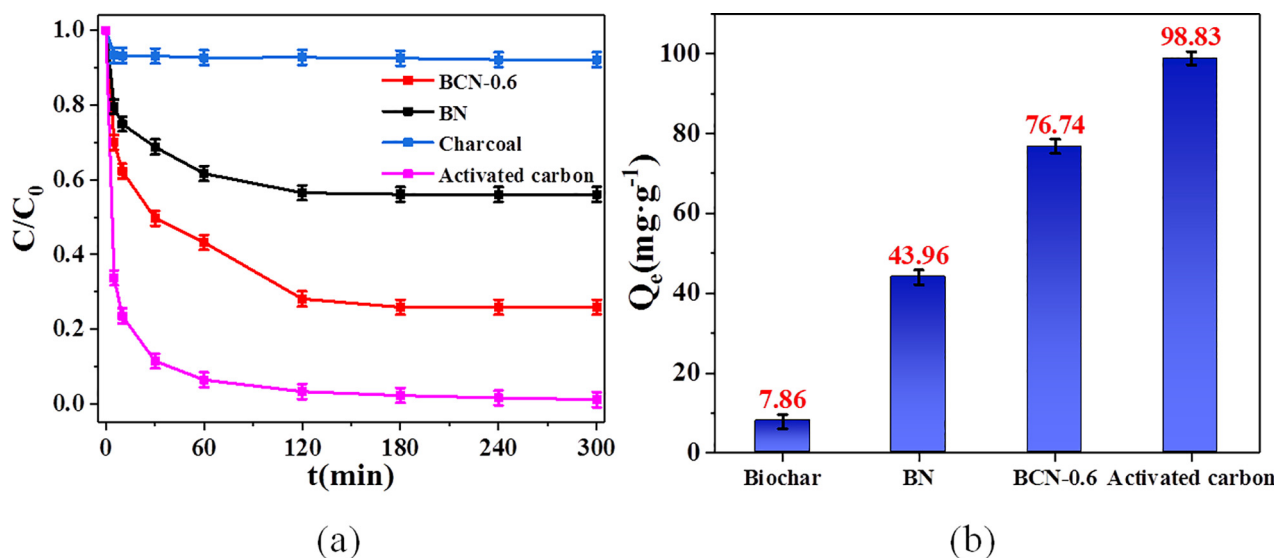


Fig. 10. (a) adsorption curves of different adsorbents; (b) the adsorption capacity of different adsorbents to TC.

3.7. Compared to other adsorbents

For better evaluating the adsorption performance of BCN-0.6, the adsorption performances of other adsorbents for TC have also been investigated. From Fig. 10, the TC adsorption capacity of biochar, BN, BCN-0.6, activated carbon and are $7.86 \text{ mg}\cdot\text{g}^{-1}$, $43.96 \text{ mg}\cdot\text{g}^{-1}$, $76.74 \text{ mg}\cdot\text{g}^{-1}$ and $98.83 \text{ mg}\cdot\text{g}^{-1}$ respectively. It is obvious that the adsorption performance of BCN-0.6 is better than that of biochar and BN. It seems that the adsorption performance of BCN-0.6 for TC is weaker than that of activated carbon. But this kind of evaluation based on the adsorption quantity

is not accurate enough since the specific surface area of BCN-0.6 is just $18.708 \text{ m}^2\cdot\text{g}^{-1}$, whereas the specific surface area of activated carbon is $607.719 \text{ m}^2\cdot\text{g}^{-1}$ (Table S7). So it is not easy to compare their adsorption performance for TC via the adsorption quantity parameter.

For better evaluating the adsorption performance of BCN-0.6 and activated carbon for TC, the adsorption quantity on per square meter have been calculated, in which it is $4.102 \text{ mg}\cdot\text{m}^{-2}$ for BCN-0.6, and $0.163 \text{ mg}\cdot\text{m}^{-2}$ for activated carbon. Therefore, the per unit specific surface adsorption capacity of BCN-0.6 for TC is 25 times higher than that of activated carbon for TC. This result suggests that the adsorption performance of BCN-0.6 for TC is better than that of activated carbon. For confirming this inference, a new LBCN-0.6 sample with surface area ($65.248 \text{ m}^2\cdot\text{g}^{-1}$) is synthesized with the same procedure to synthesize BCN-0.6 except for adding NH_4CO_3 as pore-forming agent. The surface area ($65.248 \text{ m}^2\cdot\text{g}^{-1}$) of LBCN-0.6 is smaller than that ($607.719 \text{ m}^2\cdot\text{g}^{-1}$) of activated carbon, but its adsorption capacity for TC is $118.46 \text{ mg}\cdot\text{g}^{-1}$, which is higher than that ($98.83 \text{ mg}\cdot\text{g}^{-1}$) of activated carbon. Thus, BCN-0.6 is a good adsorbent and will have potential application in the purification of wastewater.

In addition, the adsorption performance of BCN-0.6 for TC has also been compared with these reported BN materials. The porous hexagonal BN (p-BN) prepared in an inert atmosphere has a high surface area up to $1062.88 \text{ m}^2\cdot\text{g}^{-1}$ and the adsorption for TC is $128.89 \text{ mg}\cdot\text{g}^{-1}$ at the concentration of $50 \text{ mg}\cdot\text{L}^{-1}$ [69]; The boron nitride nanosheets (BNNs) also has a large specific surface area ($1801.9 \text{ m}^2\cdot\text{g}^{-1}$) and the adsorption capacity for TC is $168.74 \text{ mg}\cdot\text{g}^{-1}$ at the concentration of $50 \text{ mg}\cdot\text{L}^{-1}$ [70]. While BCN-0.6 has a surface area ($18.708 \text{ m}^2\cdot\text{g}^{-1}$) and the adsorption capacity for TC is $91.56 \text{ mg}\cdot\text{g}^{-1}$ at the concentration of $50 \text{ mg}\cdot\text{L}^{-1}$ (Fig. 5b). The adsorption quantity on per square meter of p-BN and BNNs for TC is $0.121 \text{ mg}\cdot\text{m}^{-2}$ and $0.094 \text{ mg}\cdot\text{m}^{-2}$ respectively, which are all lower than that ($4.894 \text{ mg}\cdot\text{m}^{-2}$) of BCN-0.6. The good adsorption performance of BCN-0.6 is most likely due to that these

oxygen-containing functional groups on the surface of BCN-0.6, which is favorable for the formation of hydrogen bonding with TC. These oxygen-containing functional groups on BCN-0.6 surface are from that BCN-0.6 is synthesized via oxygen-limited method [20,46,71].

4. Conclusion

A series of carbon-doped boron nitride materials (BCN) have been first time successfully synthesized via oxygen-limited method, in which BCN-0.6 has the largest adsorption capacity ($76.74 \text{ mg}\cdot\text{g}^{-1}$) for TC among all of the BCN samples. The adsorption process of BCN-0.6 for TC follows the Langmuir model and pseudo-second-order model, and the adsorption process is endothermic. BCN-0.6 can increase the pH values of TC solution to 7.87 from 3.80, while BN can just increase the pH to 4.12 from 3.80. This is mostly due to that doping of carbon into boron nitride skeleton weakens the conjugation between N and B atoms in BN and makes it capture more H^+ than BN. The adsorption capacity of BCN-0.6 for TC is $76.74 \text{ mg}\cdot\text{g}^{-1}$ at pH 8.87, being higher than that ($63.08 \text{ mg}\cdot\text{g}^{-1}$) of BCN-0.6 for TC at pH 3.80. Experimental and theoretical studies suggest that hydrogen bonding between functional groups in TC and BCN-0.6 are the main driving force for TC adsorption by BCN-0.6. During the adsorption process, the captured proton by BCN-0.6 can transfer to TC, which makes that TC become positive and BCN become negative. This will increase the electrostatic interaction between TC and BCN-0.6. In addition, $\pi\cdots\pi$ stacking between the aromatic ring of TC and the surface of BCN-0.6 will also make contribution for TC adsorption by BCN-0.6. The adsorption properties of BCN-0.6 for TC can be easily regenerated by washing with water. Furthermore, the per unit specific surface adsorption capacity ($4.102 \text{ mg}\cdot\text{m}^{-2}$) of BCN-0.6 for TC is far higher than that ($0.163 \text{ mg}\cdot\text{m}^{-2}$) of commercial activated carbon for TC. Thus, BCN-0.6 is a good adsorbent for water purification.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

The images of BCN samples; the UV spectra of TC at different pH values; the standard curve of TC; The UV spectra of TC solution and TC solution filtered by 0.22 μm membrane; the molecular structure of BN model; the molecular structure of BCN model; the molecular structure of TC under neutral conditions, in which a proton is lost from the O atom with a red circle; The pseudo-first-order kinetic adsorption of TC by BCN-0.6; the pseudo-second-order kinetic adsorption of TC by BCN-0.6 etc. are provided in supporting information. Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2020.124136>.

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